

Tawaun A. Lucas, Ph.D.

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Accomplished computational biologist with a Ph.D. from Stanford, specializing in target discovery and biomarker development through the application of AI/ML to high-dimensional multi-omic data. Expert in building scalable computational workflows to integrate single-cell and spatial transcriptomics for disease indications in neurodegeneration and immunology. Proven track record in bridging the gap between dry-lab innovation and wet-lab validation to accelerate therapeutic R&D pipelines.

Education

Stanford University School of Medicine
Ph.D. in Neurosciences

Sept. 2014 – May 2021

- **Thesis advisor:** Marion Buckwalter
- **Relevant Coursework** Molecular and Cellular Neurobiology, Molecular and Cellular Immunology, Algorithms for Computational Biology, Computational Neuroscience

California State University, Northridge
B.A in Psychology (Honors)

Aug. 2009 – May 2014

- Student-Athlete (Track and Field)
- **Relevant Coursework** Human Genetics, Psychobiology, Advanced Statistics

Work Experience

Senior Computational Biologist
Chan Zuckerberg Biohub

Redwood City, CA
Nov. 2025 – Present

- Leading the development of [AI workspace](#), a no-code web platform that streamlines scRNA-seq analysis through automated workflows, integrated AI models, hosted computing, and interactive visual interfaces.
- Spearheading benchmarking and evaluation studies for AI/ML foundation models to ensure scientific rigor in target identification.
- Collaborating with cross-functional teams to build open-source scientific products that enable the broader research community to leverage multimodal single-cell and spatial data.

Computational Biologist (Contract)
Genentech Inc.

South San Francisco, CA
Aug. 2025 – Present

- Collaborate within cross-functional teams to support immunology target discovery and therapeutic development for Inflammatory Bowel Disease (IBD).
- Develop computational frameworks for the integration and analysis of multimodal data (molecular, cellular, clinical, and imaging) to enhance R&D decision-making.
- Utilize single-cell and spatial transcriptomics, histology, and foundation models for target prioritization and biomarker discovery/development.
- Identified and brought forward two targets for prioritization and created workstreams that reduced dataset analysis time by approximately 10%.

Postdoctoral Fellow
Genentech, Inc.

South San Francisco, CA
April 2021 – August 2025

- Created a large-scale single-cell transcriptomic atlas of human and mouse astrocytes across neurodegenerative diseases to inform foundational models in biomarker discovery.
- Led the development of analysis pipelines for spatial transcriptomics datasets (e.g., 10x Visium, 10x Xenium) and implemented machine learning algorithms to integrate gene expression and spatial location, generating insights about subclasses of glia and fibroblasts in neurodegenerative and inflammatory diseases.
- Generated experimental frameworks unifying multi-omics sequencing with CRISPR and ATAC (Perturb-seq, CITE-seq, 10x Epi Multiome) analysis, generating novel targets in inflammatory diseases.
- Seamlessly interfaced between computational and wet-lab scientists, reciprocally to allow for a lab-in-loop integration, easing the ability of our team to drive experimentation and analysis promptly.

- Collaborated and guided 3 junior researchers, 1 research associate, and 2 interns as well as specialists across neuroscience and immunology to address pertinent biological and clinical questions of interest.

Graduate Researcher

Buckwalter Lab- Neurology (Stanford)

Stanford, CA

Aug. 2014 – Feb. 2021

- Investigated peripheral glia in the spleen and their role in neuroimmune communication to address pertinent questions about immunity after head injury.
- Designed and deployed RNA-seq analysis frameworks and workflows using Snakemake and contributed to developing intricate transcriptomic studies, delivering insights into neuroimmune connections within the spleen and gut.
- Developed and maintained “Shiny” R apps for real-time data visualization, enhancing accessibility and utility for academic research labs

Summer Researcher

McClure Lab- (Stanford)

Stanford, CA

June 2013 – Aug. 2013

- Investigating the regulatory role of GABA on human action selection and decision making.
- Generated tasks to assess aspects of decision-making using MATLAB
- Conducted and analyzed large-scale fMRI scans

Undergraduate Researcher

Tonyan Lab - (Northridge)

Northridge, CA

October 2012 – May 2014

- Investigating the effects of chaotic environments on child development and decision making.

Publications

SpatialAgent: An Autonomous AI Agent for Spatial Biology March 2025

Wang, H, He, Y, Coelho, P, Bucci, M..., **Lucas, T A**..., Rozenblatt-Rosen, O, Leskovec, J, Regev, A

Submitted. *Preprint: doi:10.1101/2025.04.03.646459v1* [🔗](#)

Integrated Cross-Disease Atlas of Human and Mouse Astrocytes Reveals Heterogeneity and Conservation of Astrocyte Subtypes in Disease Feb. 2025

Lucas, T A, Novikova, G, Rao, S, Wang, Y, Laufer, B., Pandey, S, Webb, M, Jorstad, N, Friedman, B, Hanson, J, Kaminker, J

In Revision-Nature Neuroscience *Preprint: doi:10.1101/2025.02.12.637903* [🔗](#)

Translatome analysis reveals microglia and astrocytes to be distinct regulators of inflammation in the hyperacute and acute phases after stroke Aug 2023

Hernandez, V, Lechtenberg, K J, Perterson, T, Zhu, L, **Lucas, T A** ... Bukwalter, M
Glia. *doi:10.1002/glia.24377* [🔗](#)

An RNA-sequencing transcriptome of the rodent Schwann cell response to peripheral nerve injury April 2022

Brosius-Lutz, A, **Lucas, T A**, Carson, G, Caneda, C, Barres, B, Bukwalter, M, Sloan, S
Journal of Neuroinflammation. *doi:10.1186/s12974-022-02462-6* [🔗](#)

Spleen glia are a transcriptionally unique glial subtype interposed between immune cells and sympathetic axons March 2021

Lucas, T A, Zhu, L, Bukwalter, M

Glia. *doi:10.1002/glia.23993* [🔗](#)

Obesity drives delayed infarct expansion, inflammation, and distinct gene networks in a mouse stroke model June 2020

Perterson, T, Lechtenberg, K J, **Lucas, T A**, ... Bukwalter, M

Translational Stroke Research. *doi:10.1007/s12975-020-00826-9* [🔗](#)

Technical Skills

Languages: R, Linux/Bash, Python, Matlab

Computational Analysis: scRNA-seq, scATAC-seq, CITE-seq, Perturb/CRISPR/CROP-seq, 10x Multiome, 10x Visium & Xenium, MERFISH, Cell Atlases, Tidyverse, Numpy, Matplotlib, Pandas, Version control (git)

Computational Technologies: High-Performance/Cloud Computing (HPC/AWS), Machine Learning, Shiny

Machine Learning Frameworks: PyTorch, Scikit-learn, TensorFlow

Teaching & Service

Reviewer for Scholarly Journals:

- Nature Neuroscience
- Journal of Clinical Investigation
- Glia

Program Manager/Teacher

Apogee Neuroscience Camp

Stanford, CA

Apr. 2018 – June 2020

Apogee Summer Camp is a three-week camp that educates senior high school students on several neuroscience topics. As program manager, I recruited students and teachers and managed the program's logistics, which included 50 students. As an instructor, I designed and implemented lesson plans focused on cognition and behavior. On average, this involved a commitment of 15 hours a week.

Brain Day Teacher

Stanford Neurosciences Program

Stanford, CA

Feb. 2015 – April 2021

As a Neuroscience Educator, I traveled to middle and high schools in the Palo Alto and East Palo Alto areas to teach one-hour interactive classes on the brain and its functions. Working collaboratively in groups of three, we designed engaging curricula and led dynamic activities. This program aimed to spark students' curiosity about the brain and neuroscience, fostering a more profound interest in the subject.

Senior Program Leader

Stanford ADVANCE Summer Institute

Stanford, CA

Mar. 2018 – Aug 2018

The ADVANCE summer institute is a 10-week bridge program for incoming Ph.D. students at the Stanford School of Medicine. As a Senior Program Leader, I was one of a two-person team that coordinated the program and organized the students. I taught weekly personal development seminars and recruited students into the program. On average, this was a 15-hour-a-week commitment

Founder

Diversity Center of Representation and Empowerment

Stanford, CA

Oct. 2017

The DCORE space is a physical space at the Stanford School of Medicine that engages underrepresented populations and provides a safe space for the community to gather to host events, study, or invite guest speakers to campus for seminars. As a founding member, I negotiated with the administration to secure this space. Press release: <https://stan.md/2ImI4fx> 

Senior Program Leader

Stanford Summer Research Program

Stanford, CA

Mar. 2016 - Aug. 2017

The SSRP is a nine-week intensive research experience for undergraduates from diverse backgrounds and skill levels nationwide. In addition to the research they conduct in labs, the program emphasizes professional development and preparation for grad school applications. As program lead, I was responsible for admissions, reading, and assessing fit for program and lab placement. I was also responsible for hiring Program Assistants to mentor undergraduates. On average, this was a 25-hour-a-week commitment.

Graduate Mentor

Earnest Houston Johnson Scholars

Stanford, CA

Dec. 2014 - Dec. 2015

This unique mentoring opportunity allowed me to mentor eight first-year Stanford University students majoring in biomedical science. We met weekly to discuss issues during their quarter, including applying to internships, building academic and professional resumes, and time management and priority setting. On

average, this was a five-hour-a-week commitment.

Tutor

Academic Services for Student Athletes-CSUN

Northridge, CA
Jan. 2010 - May 2014

Provided one-on-one and group tutoring sessions in statistics and biology for undergraduate and graduate students, developing personalized lesson plans to address individual needs. Assisted students with complex topics like probability, hypothesis testing, cell biology, and genetics, improving their academic performance and confidence through tailored teaching techniques and continuous feedback.

Awards

Stanford School of Medicine Commencement Speaker	June 2020
Diversifying Academia and Recruiting Excellence (DARE) Fellowship	Aug. 2018 - Aug. 2020
National Science Foundation-Graduate Research Fellowship (GRFP)	Apr. 2015 - Apr. 2018
Stanford Biosciences - Excellence in Diversity and Societal Citizenship	May 2015
Stanford Office of Graduate Education ADVANCE Fellow	July 2014
NIH Minority Access to Research Careers (MARC) Fellowship	May 2012 - May 2014
CSU Northridge Undergraduate University/Community Service Award	May 2014
SRCD - Frances Degen Horowitz Millennium Scholar	April 2013